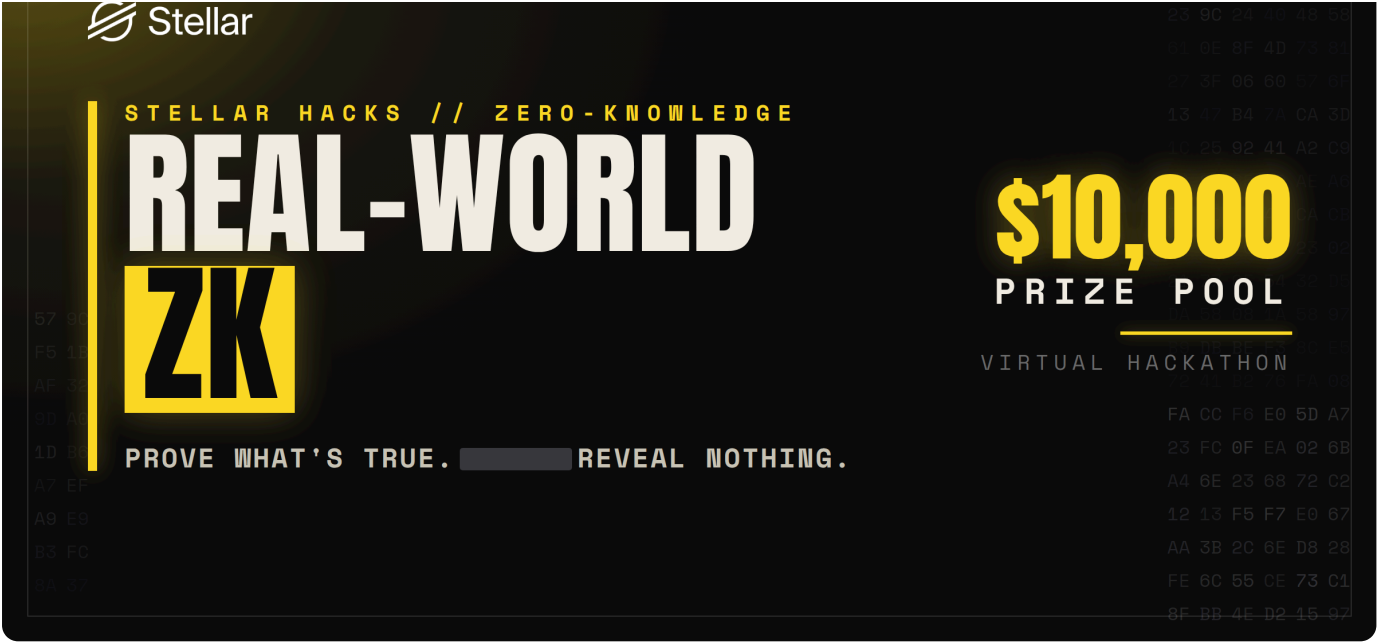






PRIZE POOL

10,000 USD



EVENT TIMELINE

9 days left for submission

Upcoming



Submission

2026/06/15 12:30

Deadline

2026/06/30 00:30



Virtual



HACKATHON TAGS

Blockchain ZK Zero Knowledge Rust Noir RISC Zero Soroban Circom



PLATFORM TECHNOLOGY

Stellar ZK



SUBMISSION REQUIREMENTS



Stellar Development Foundation



Message

Stellar Hacks: Real-World ZK

Submit BUIDL

Register as Hacker





OUTLINE

Start Here: ZK Privacy on Stellar

AI Development Assistance

On-Chain ZK Verifiers (Reference Implementations)

ZK Circuit Tooling

Further Privacy Context

Core Stellar Dev Tools

Smart Contract Building Blocks (Docs)

Community Resources

New to ZK on Stellar? Start with the two official docs pages — [Privacy on Stellar](#) and [ZK Proofs on Stellar](#) — then point your AI agent at the Stellar Skills below.

Start Here: ZK & Privacy on Stellar

- **ZK Proofs on Stellar (docs):** <https://developers.stellar.org/docs/build/apps/zk> — The core reference. Explains the BN254 and Poseidon/Poseidon2 host functions, what they do, and how proof verification works on Stellar. Includes code examples and links to circuit tooling.
- **Privacy on Stellar (docs):** <https://developers.stellar.org/docs/build/apps/privacy> — Overview of the privacy stack: Privacy Pools, Confidential Tokens, on-chain ZK verifiers, and the underlying cryptographic primitives. The best map of the whole landscape.
- **Announcing Stellar X-Ray (Protocol 25):** <https://stellar.org/blog/developers/announcing-stellar-x-ray-protocol-25> — Background on why these primitives were added and the long-term privacy strategy behind them.





ZK builders and why proof verification got cheaper.

AI Development Assistance

We assume most of you will be building with an AI agent. Give it the right Stellar context first — it dramatically improves the code it writes.

- **Stellar Skills (start here):** <https://skills.stellar.org/> — Agent-readable documentation for building on Stellar, with dedicated skills for Soroban, dApps/wallets, assets, data/APIs, agentic payments, and **ZK Proofs**. Works with any AI agent. The simplest path: tell your agent *"Read skills.stellar.org before you start building on Stellar."*
- **ZK Proofs skill (direct):** <https://skills.stellar.org/skills/zk-proofs/SKILL.md> — Verify Groth16 zero-knowledge proofs on Stellar using BLS12-381, BN254, and Poseidon primitives.
- **Stellar Dev Skill (repo):** <https://github.com/stellar/stellar-dev-skill> — The underlying open-source skill repo. Covers Soroban, SDKs, RPC, wallet integration, passkeys, and security patterns.

Install in Claude Code: `/plugin marketplace add stellar/stellar-dev-skill` then `/plugin install stellar-dev@stellar-dev` **Cursor:** add `stellar/stellar-dev-skill` **Codex:**

`git clone [https://github.com/stellar/stellar-dev-skill](https://github.com/s`

- **stellar-build:** <https://github.com/kaankacar/stellar-build> — Stellar development journey installer of 42 skills covering the full journey from idea to mainnet deploy and SCF grant submission, with six DevRel-persona agents.
- **OpenZeppelin Skills:** <https://github.com/OpenZeppelin/openzeppelin-skills> — Claude Code skills for secure Stellar contract development. Install:
`/plugin marketplace add OpenZeppelin/openzeppelin-skills` and
`/plugin install openzeppelin-skills`
- **Building with AI (docs):** <https://developers.stellar.org/docs/build/building-with-ai> — Stellar's guidance on AI-assisted development.





On-Chain ZK Verifiers (Reference Implementations)

These are the closest thing to "starter code" for this hackathon — deployable verifier contracts you can study, fork, and build on.

- **RISC Zero (Groth16) verifier:** <https://github.com/NethermindEth/stellar-risc0-verifier> — Verifies Groth16 proofs created with RISC Zero's zkVM (write your provable program in Rust). Built by Nethermind's ZK team. Companion article: <https://stellar.org/blog/developers/risc-zero-verifier>
- **UltraHonk verifier (Noir / Barretenberg):**
- <https://github.com/yugocabrio/rs-soroban-ultrahonk>
- https://github.com/indextree/ultrahonk_soroban_contract — Verifier for circuits built with Aztec's Noir language. A clean pattern for proving valid solutions/state without revealing them.
- **Stellar Private Payments (Privacy Pools PoC):**
<https://github.com/NethermindEth/stellar-private-payments> — Nethermind's proof-of-concept Privacy Pools implementation using Circom circuits, Groth16 proofs, and Stellar smart contracts. Includes a pool contract, on-chain Groth16 verifier, and ASP membership/non-membership contracts. Proofs are generated client-side in the browser via WebAssembly, so secrets never leave the device. Docs: <https://nethermindeth.github.io/stellar-private-payments/>
- *Caution: research prototype, not audited. Don't use with real assets.*

ZK Circuit Tooling

- **Noir (Aztec):** <https://noir-lang.org/docs/> — A friendly, Rust-like DSL for writing ZK circuits. Pairs with the UltraHonk verifier above. Protocol 26 made Noir proof verification on Stellar significantly cheaper.
- **RISC Zero (zkVM):** <https://dev.risczero.com/> — Write your provable program in ordinary Rust and prove its execution. Pairs with the RISC Zero verifier above.



- Soroban SDK — BN254 docs: https://docs.rs/soroban-sdk/latest/soroban_sdk/migrating/v25_bn254/index.html
- Soroban SDK — Poseidon docs: https://docs.rs/soroban-sdk/latest/soroban_sdk/migrating/v25_poseidon/index.html
- Protocol CAPs (deep cuts): BN254 — [CAP-0074](#) · Poseidon/Poseidon2 — [CAP-0075](#) · BLS12-381 — [CAP-0059](#)
- Soroban P25 preview examples: <https://github.com/jayz22/soroban-examples/tree/p25-preview/p25-preview>

Further Privacy Context

- Confidential Token Association: <https://www.confidentialetoken.org/> — Open standard (SDF, Nethermind, OpenZeppelin, Zama) for encryption-based on-chain confidentiality compatible with existing token interfaces. Overview/demo video: <https://www.youtube.com/watch?v=6NnDqVQYOHM>
- Privacy Pools whitepaper (Buterin, Illum, Nadler, Schär, Soleimani): <https://privacypools.com/whitepaper.pdf> — The conceptual basis for compliant privacy pools (deposits/withdrawals visible, in-pool transfers private, with ASP allow/deny lists).

Core Stellar Dev Tools

- Stellar Docs: <https://developers.stellar.org/> — Core documentation for building on Stellar.
- SDKs: <https://developers.stellar.org/docs/tools/sdks> — Libraries to interact with the network in your preferred language. (Use the latest SDK version for Protocol 26 support.)
- Stellar CLI: <https://developers.stellar.org/docs/tools/cli> — Build, deploy, and interact with Soroban smart contracts from the command line.





testnet accounts).

- **Quickstart:** <https://developers.stellar.org/docs/tools/quickstart> — Run a local Stellar network via Docker for development/testing.
- **Scaffold Stellar:** <https://scaffoldstellar.org> — CLI for the full Stellar app development lifecycle: contract management, testing, and deployment with best practices baked in.
- **Stellar Wallets Kit:** <https://stellarwalletskit.dev/> — Plug-and-play wallet connections with a unified API.
- **OpenZeppelin on Stellar:** <https://www.openzeppelin.com/networks/stellar> — Audited contract library, Contracts Wizard, Contracts MCP server, Relayer, and Soroban security detectors.

Smart Contract Building Blocks (Docs)

- **Smart Contracts — Getting Started:** <https://developers.stellar.org/docs/build/smart-contracts/getting-started>
- **Contract Authorization:** <https://developers.stellar.org/docs/build/guides/auth>
- **Contract Storage:** <https://developers.stellar.org/docs/build/guides/storage>
- **Contract Testing:** <https://developers.stellar.org/docs/build/guides/testing>

Community Resources

- **Stellar Ecosystem Resources:** <https://github.com/stellar/ecosystem-resources/> — Workshop activations and reference guides for Soroban, wallets, DeFi protocols, OpenZeppelin, tokens, and security.
- **Stellar Hackathon FAQ:** <https://github.com/briwylde08/stellar-hackathon-faq> — Community-compiled FAQ from building on Stellar.





building from scratch.



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